

Trainings ▾

General Information

All Applications

Note : This procedure applies to Stage 1 filters on engines with electronically actuated injectors only.

Use the following procedure for engines with mechanically actuated injectors. Refer to Procedure 006-015 in Section 6. (</qs3/pubsys2/xml/en/procedures/56/56-006-015-tr.html>)

Use the following procedure for Stage 2 fuel filters. Refer to Procedure 006-076 in Section 6.

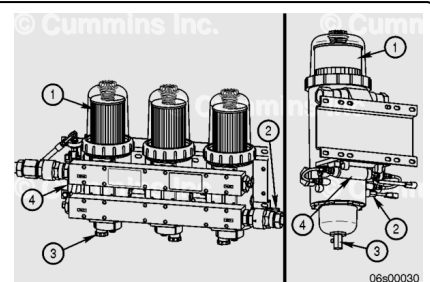
(</qs3/pubsys2/xml/en/procedures/56/56-006-076-tr.html>)

The Stage 1 filters are mounted off the engine and use seven micron filtration media. The fuel priming pump is mounted with the Stage 1 filter assembly. The Stage 1 filters are mounted at the original equipment manufacturer (OEM) discretion, provided Cummins Inc. application guidelines are met.

The FH234 series is on the left of the illustration and the FH239 series is on the right. The filter consists of the following components:

1. Stage 1 filter
2. Water in fuel sensor
3. Drain valve
4. Lift pump.

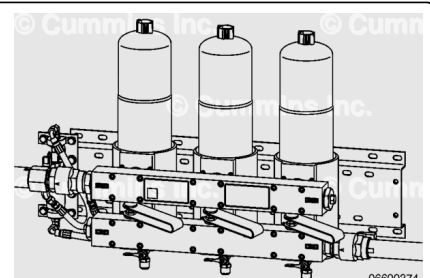
Note : The following steps show the FH234 series, but the FH239 series steps are very similar.



Marine Applications

Marine applications use either an FH234 Sea Pro™ spin-on style filter or an FH240 Sea Pro™ cartridge style filter.

Marine applications use a filter head that utilizes valves to allow for changing of the filters while the engine is in operation.



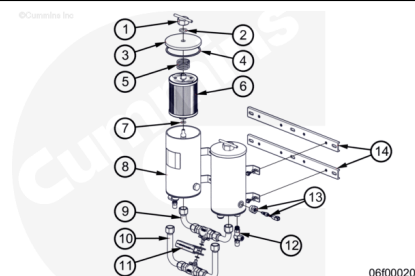
The valves are located between the inlet and outlet fuel manifolds on the filter assembly. They consist of a lever type valve with an ON or OFF position.

FH234 Stage 1 filters are mounted off the engine and use seven micron filtration media. The fuel priming pump is mounted with the Stage 1 filter assembly. The Stage 1 filters are mounted at the OEM discretion, provided Cummins Inc. application guidelines are met.

FH240 Stage 1 filters are mounted off engine and use five micron filtration media. The fuel priming pump is mounted on-engine. The Stage 1 filters are mounted at OEM discretion, provided Cummins Inc. application guidelines are met.

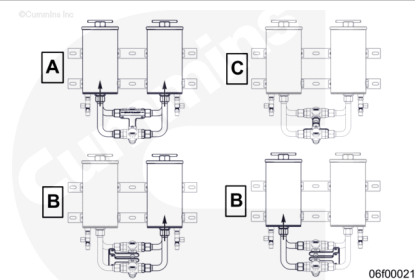
The Sea Pro™ FH240 Classed filter consists of the following:

1. Nut
2. Nut Seal
3. Service Cap
4. Service Cap Seal
5. Spring
6. Stage 1 Filter Element and Seals (Includes 2, 4, and 7)
7. Stand Pipe Gasket
8. Filter Housing
9. Fuel Out Manifold
10. Fuel In Manifold
11. Handle
12. Drain Valve
13. Water in fuel Sensor
14. Mounting Bracket.



Valve Operation - Cartridge Filters

- For dual unit flow, the valve handle should be facing upward.
- For single unit flow, rotate the valve handle either left or right 90 degrees depending on which unit is to be used.
- To shut off flow, rotate the valve handle 180 degrees from the dual flow position.



Preparatory Steps

Industrial Applications

⚠ WARNING ⚠

Depending on the circumstance, diesel fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

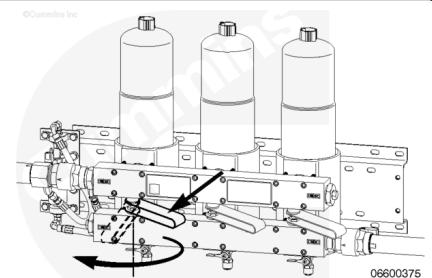
Stage 1 Filters:

- Disconnect the batteries. See equipment manufacturer service information.
- Close the fuel supply valve.

Marine Applications

⚠ WARNING ⚠

Depending on the circumstance, diesel fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.



Stage 1 Filters:

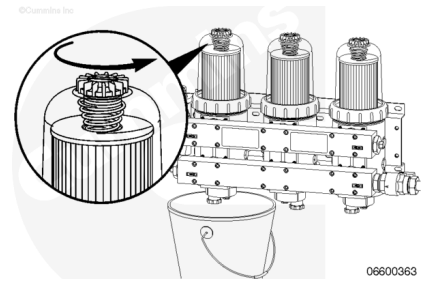
- With the engine operating, move the selected valve handle to the OFF position (depending on which filter is to be changed).

Remove

Industrial Applications

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

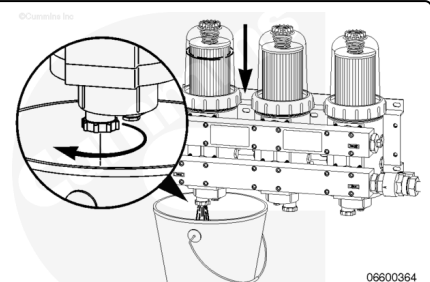


Stage 1 Filters:

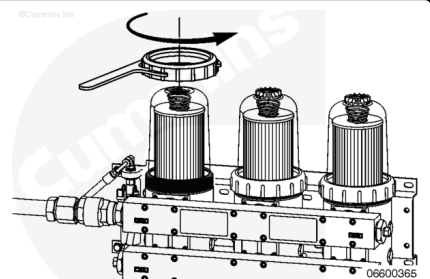
Place a suitable container under the fuel filter to be replaced.

Use a collar/vent cap wrench, Part Number 3944451 S, to open the vent cap to break the air lock in the filter.

Open the fuel drain valve and allow the fuel level to drain to a point below the collar. Close the drain valve.



Use the collar/vent cap wrench, Part Number 4944451 S, to loosen the collar.

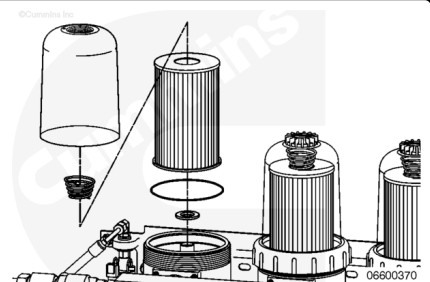


Remove the clear cover, filter spring, fuel filter element, and o-ring.

Discard the o-ring.

Remove the sealing grommet from the filter stud and discard.

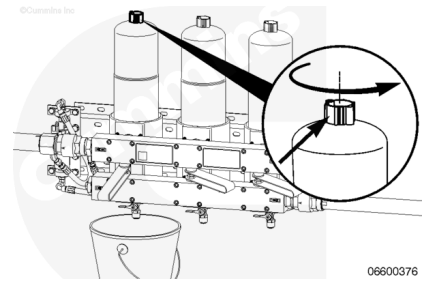
Note : The FH239 series has two filter elements that are stacked together. Make sure both elements are removed and replaced at the same time.



Marine Applications

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.

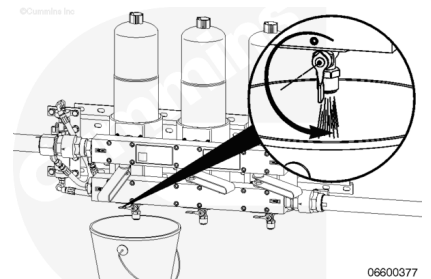
Stage 1 Spin-On Filters:

Place a suitable container under the fuel filter to be replaced.

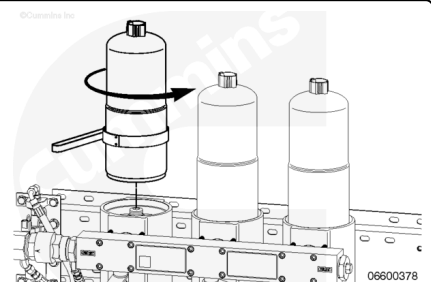
Open the vent cap to break the air lock in the filter.

Open the fuel drain valve and allow the filter to drain.

Close the drain valve when the drainage slows to a trickle.

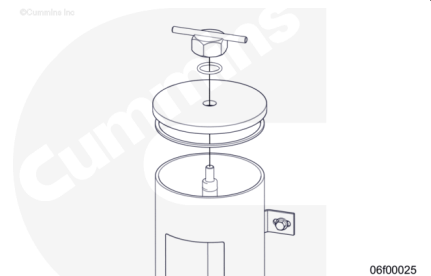


Remove the Stage 1 fuel filters with filter wrench, Part Number 3400157, or equivalent.



⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.



⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. The fuel must be disposed of in accordance with local environmental regulations.

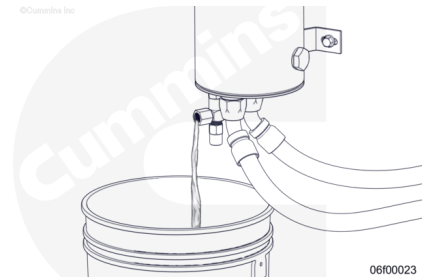
Stage 1 Cartridge Filters:

Place a suitable container under the fuel filter to be replaced.

Remove the service cap to break the air lock in the filter.

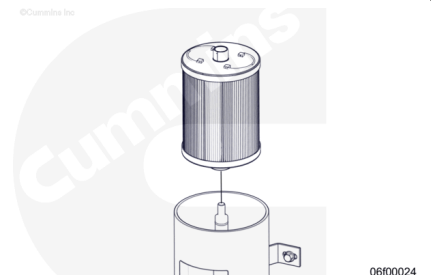
Open the fuel drain valve and allow the filter to drain.

Close the drain valve when drainage slows to a trickle.



Remove the Stage 1 fuel filter.

After the filter element has been removed, make sure the selector valve remains in the OFF position.

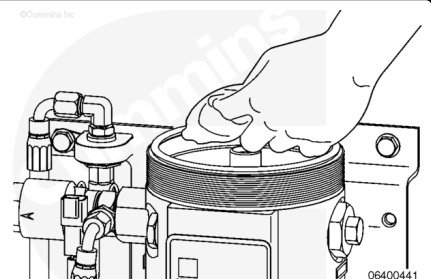


Clean

All Applications

Note : The Industrial application filter is shown. Although different in appearance, the procedure remains the same for Industrial and Marine applications.

Check the sealing surface of the filter body. Wipe the surface with a clean, lint-free cloth to remove any debris or dirt.



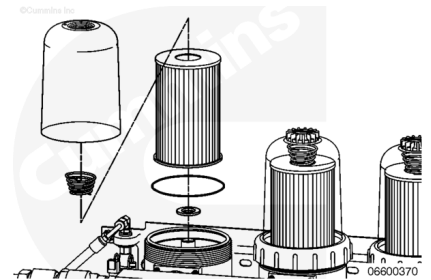
Install

Industrial Applications

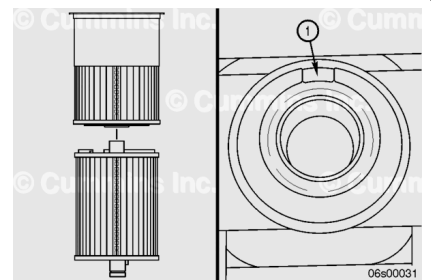
Stage 1 Filters:

Note : Verify the used o-ring has been removed and discarded before installing the new o-ring.

Install a new o-ring, filter element (supplied with a sealing grommet inserted into the filter element), filter spring, and clear cover.

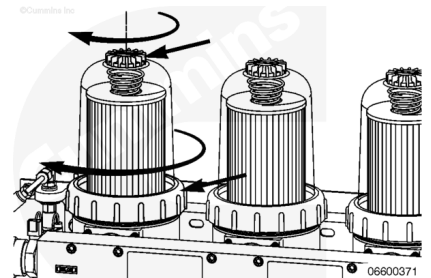


The FH239 Series has two filter elements that are stacked together. The bottom element has a key that **must** match the keyway (1) in the filter head.



Install the vent cap and collar onto the clear cover and hand tighten **only**.

Do **not** use tools to tighten the collar.

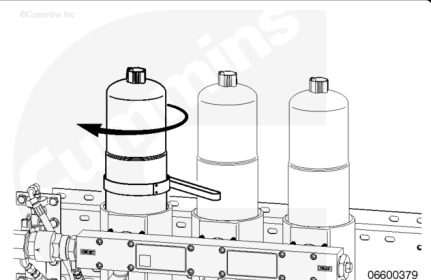


Marine Applications

Stage 1 Spin-On Filters:

Install the Stage 1 filter on the filter head. Turn the filter until the gasket touches the surface of the filter head.

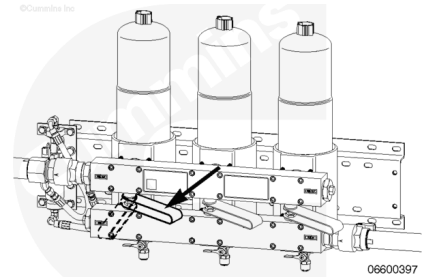
Tighten the filter an additional $\frac{1}{2}$ to $\frac{3}{4}$ turn.



Move the valve handle to the ON position.

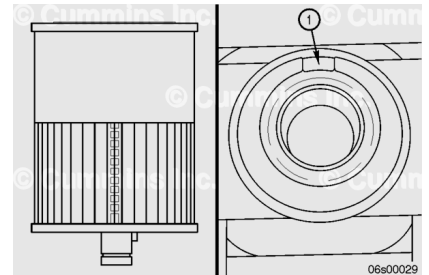
Raise the engine rpm for 1 minute to purge air from the system.

Repeat the procedure for the remaining filter replacements.

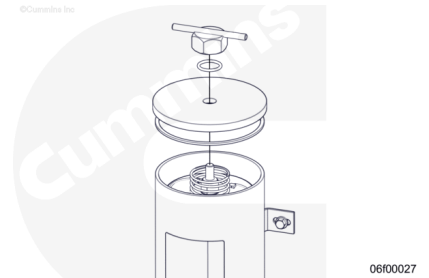


Stage 1 Cartridge Filters:

Install a new pre-lubricated o-ring on the filter element. Insert filter element into the filter housing. The FH240 series element has a key that **must** match the keyway (1) in the filter head.



Place service cap on top of filter housing with replacement service cap seal, making sure cap is centered on housing. Install knob and hand-tighten.



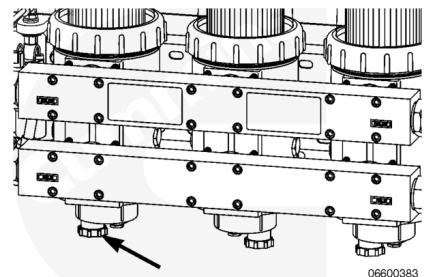
Prime

Industrial Applications

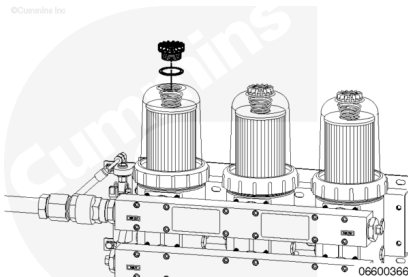
Stage 1 Filters:

Check that the drain valve is closed at the base of the filter.

Close the fuel supply valve.



Use the collar/vent cap wrench, Part Number 3944451 S, to remove the vent cap from the top of the clear cover.

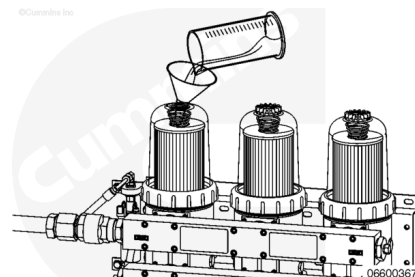


Fill the clear cover with enough clean diesel fuel to cover the bottom half of the fuel filter.

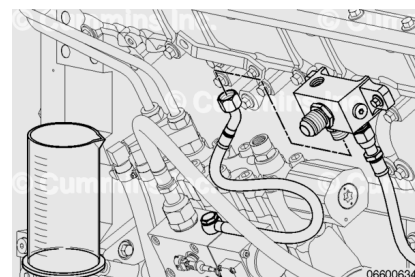
Install the vent cap.

Tighten the vent cap by hand **only**.

Open the fuel shutoff valve.



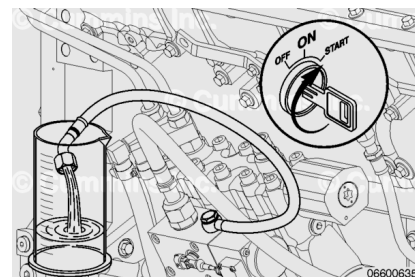
Remove the air bleed line from the drain manifold block and place it into a collection container.



Turn the keyswitch ON and observe fuel exiting the air bleed line.

When a steady stream of fuel is observed, turn the keyswitch OFF.

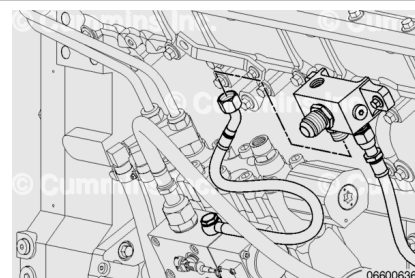
If the priming pump stops operating before a steady stream of fuel is observed, turn the keyswitch OFF, wait 30 seconds, and turn the keyswitch back ON. Repeat as necessary to observe a steady stream of fuel.



Install an air bleed line to drain the manifold block.

Torque Value:

Air bleed line 45 n•m [33 ft-lb]



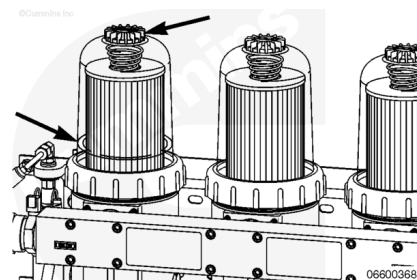
Start the engine.

When the lubrication system reaches its normal operating pressure, increase the engine speed to high idle for 1 to 2 minutes.

Once all the air is purged, loosen the vent cap until the fuel level lowers to just above the collar.

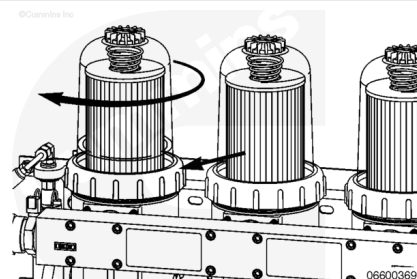
Tighten the vent cap by hand **only**.

Note : The clear filter cover will **not** fill completely during engine operation. It will fill over time and the fuel level will rise as the filter element traps contaminants.



Tighten the collar a second time while the engine is operating.

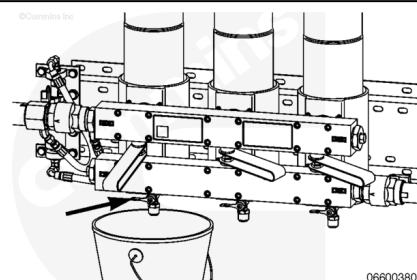
Do **not** use any tools to tighten the collar or damage can possibly result.



Marine Applications

⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.

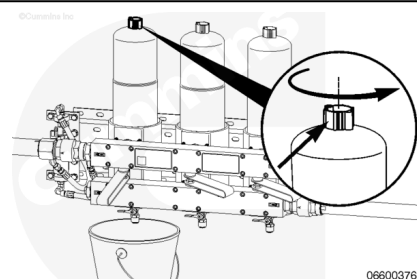


Stage 1 Spin-On Filters:

Check that the drain valve is closed at the base of the filter.

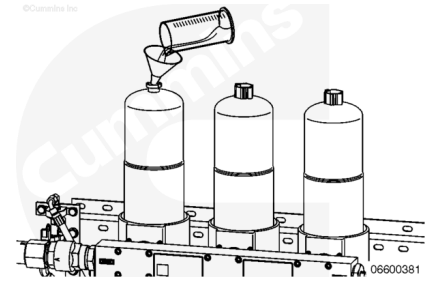
Close the fuel supply valve.

Remove the vent cap from the top of the cover.



⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.



Fill the filter with clean diesel fuel.

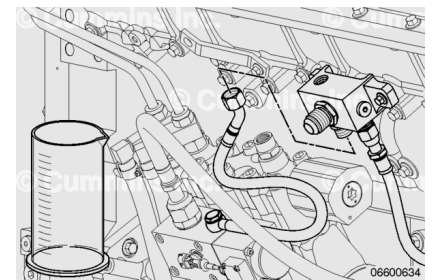
Install the vent cap.

Tighten the vent cap by hand **only**.

Open the fuel shutoff valve.

Stage 2 Filters:

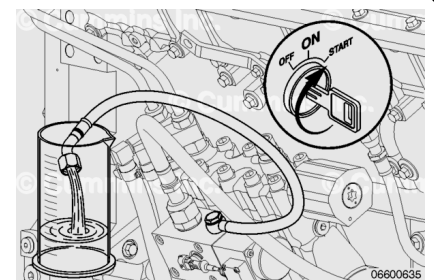
Remove the air bleed line from the drain manifold block and place it into a collection container



Turn the keyswitch ON and observe fuel exiting the air bleed line.

When a steady stream of fuel is observed, turn the keyswitch OFF.

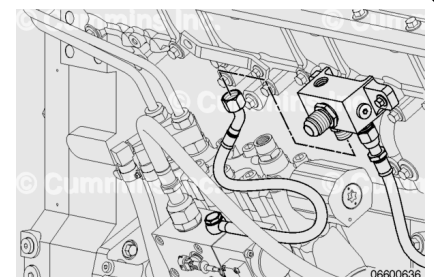
If the priming pump stops operating before a steady stream of fuel is observed, turn the keyswitch OFF, wait 30 seconds, and turn the keyswitch back ON. Repeat as necessary to observe a steady stream of fuel.



Install an air bleed line to drain the manifold block.

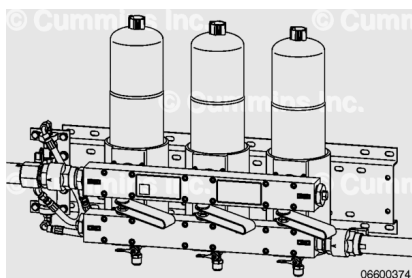
Torque Value:

Air bleed line 45 n•m [33 ft-lb]



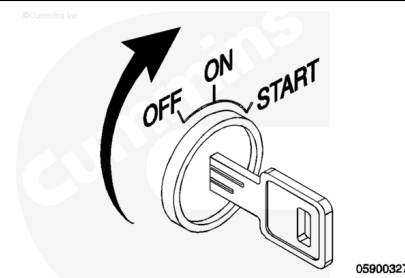
Start the engine.

When the lubrication system reaches its normal operating pressure, increase the engine speed to high idle for 1 to 2 minutes.



Stage 1 Cartridge Filters:

- Prime the fuel system using the electric priming pump.



Finishing Steps

Industrial Applications

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

⚠ WARNING ⚠

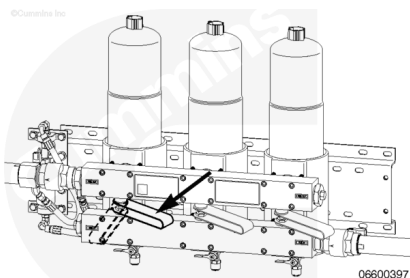
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Open the fuel supply valve.
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine and check for leaks.

Marine Applications

Move the valve handle to the ON position.

Check the filter unit selected for normal operation.

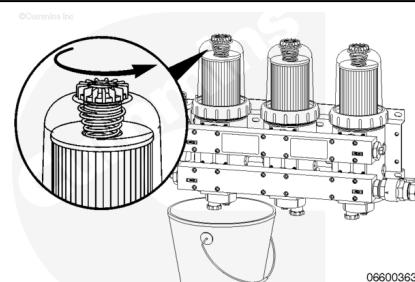


Drain

Industrial Applications

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



Stage 1 Filters:

Note : The accumulated water must be drained at least weekly. For fuel that has high water content, more frequent draining will be necessary.

To drain the accumulated water, shut the engine down.

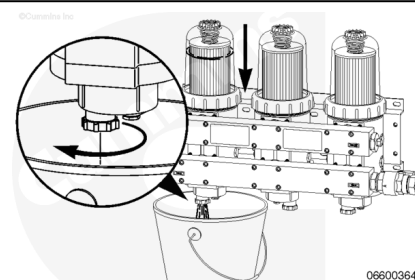
Place a suitable container under the fuel filter.

With the fuel supply valve closed, open the vent cap to break the air lock in the fuel filter.

Open the drain valve. If there is accumulated water in the system, the water will flow out first.

When fuel begins to flow out of the drain, close the drain valve.

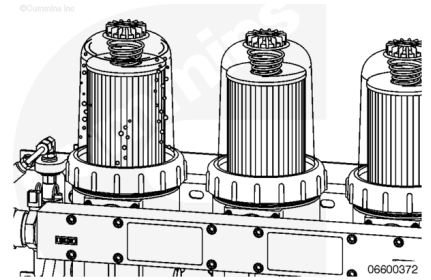
Note : Dispose of the drained fluids in accordance with local environmental regulations.



Close the vent cap.

Start the engine, raising the rpm for 1 minute to purge the air from the system.

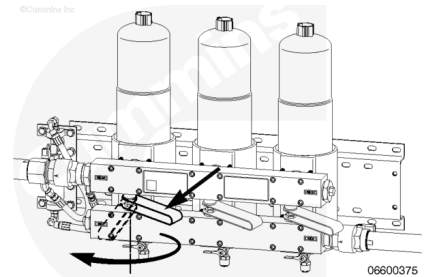
If necessary, repeat the procedure for the remaining filters to remove the excess air.



Marine Applications

⚠ WARNING ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.



Stage 1 Filters:

Note : The accumulated water must be drained at least weekly. For fuel that has high water content, more frequent draining will be necessary.

To drain the accumulated water, shut the engine down.

Place a suitable container under the fuel filter.

With the fuel filter selector valve closed, open the vent cap to break the air lock in the filter.

Open the drain valve. If there is accumulated water in the system, the water will flow out first.

When fuel begins to flow out of the drain, close the drain valve.

Note : Dispose of the drained fluids in accordance with local environmental regulations.

Close the vent cap.

Start the engine and raise the rpm for 1 minute to purge the air from the system.

If necessary, repeat the procedure for the remaining filters to remove the excess air.

